

Dendritic cells presenting tumor antigen.

PMID: 9001569 | Indexed in PubMed

Since the first identification of dendritic cells by Steinman and Cohn in 1973, progress in understanding their biology has included the development of novel methods of cell culture, recognition of critical aspects of migration and maturation, and appreciation of their major role as antigen-presenting cells (APC), and how this activity is regulated by cytokines and expression of accessory molecules. Dendritic cells are the major APC involved in the initiation of the immune response and the development of tolerance. There is considerable evidence that they can acquire antigen in the peripheral tissues and process, transport, and present it to T cells in secondary lymphoid tissue. A number of studies show that, in vitro or in vivo, antigen-pulsed dendritic cells can directly sensitize T cells and stimulate the development of antigen-specific immune responses, including both protective and therapeutic antitumor responses. In this paper, several important aspects of dendritic cell biology are discussed and a number of studies confirming the role of these professional APC in antitumor immunity are reviewed.

Exosomes derived from microRNA-540-3p overexpressing mesenchymal stem cells promote immune tolerance via the CD74/nuclear factor-kappaB pathway in cardiac allograft.

PMID: 39734479 | Indexed in PubMed

Heart transplantation is a crucial intervention for severe heart failure, yet the challenge of organ rejection is significant. Bone marrow mesenchymal stem cells (BMSCs) and their exosomes have demonstrated potential in modulating T cells, dendritic cells (DCs), and cytokines to achieve immunomodulatory effects. DCs, as key antigen-presenting cells, play a critical role in shaping immune responses by influencing T-cell activation and cytokine production. Through this modulation, BMSCs and their exosomes enhance graft tolerance and prolonging survival. To explore the immunomodulatory effects of exosomes derived from BMSCs overexpressing microRNA-540-3p (miR-540-3p) on cardiac allograft tolerance, focusing on how these exosomes modulating DCs and T cells activity through the CD74/nuclear factor-kappaB (NF- κ B) pathway. Rat models were used to assess the impact of miR-540-3p-enhanced exosomes on immune tolerance in cardiac allografts. MiR-540-3p expression was manipulated in BMSCs, and derived exosomes were collected and administered to the rat models post-heart transplantation. The study monitored expression levels of major histocompatibility complex II, CD80, CD86, and CD274 in DCs, and quantified CD4⁺ and CD8⁺ T cells, T regulatory cells, and cytokine profiles. Exosomes from miR-540-3p-overexpressing BMSCs lead to reduced expression of immune activation markers CD74 and NF- κ B p65 in DCs and T cells. Rats treated with these exosomes showed decreased inflammation and improved cardiac function, indicated by lower levels of pro-inflammatory cytokines (interleukin-1 β , interferon- γ) and higher levels of anti-inflammatory cytokines (interleukin-10, transforming growth factor β 1). Additionally, miR-540-3p skewed the profiles of DCs and T cells towards immune tolerance, increasing the ratio of T regulatory cells and shifting cytokine secretion to favor graft acceptance. Exosomes derived from BMSCs overexpressing miR-540-3p significantly enhance immune tolerance and prolong cardiac allograft survival by modulating the CD74/NF- κ B pathway, which regulates activities of DCs and T cells. These findings highlight a promising therapeutic strategy to improve heart transplantation outcomes and potentially reduce the need for prolonged immunosuppression.

Monocyte-derived dendritic cells as antigen-presenting cells in T-cell proliferation and cytokine production.

PMID: 18615247 | Indexed in PubMed

Dendritic cells (DC) are widely considered to be the major antigen-presenting cell (APC) type in immune responses. These cells are obtained from adherent cells or are purified CD14(+) monocytes from peripheral blood mononuclear cells (PBMC) by in vitro stimulation with granulocyte, macrophage-colony-stimulating factor (GM-CSF) plus interleukin (IL)-4. They express high levels of MHC class II and costimulatory molecules, internalize Ag rapidly via Fc receptors and mannose receptors, and, by macropinocytosis, produce large amounts of IL-12 on CD40 ligation, and are potent in presenting soluble Ag and in stimulating allogeneic mixed-leukocyte reactions. To study primary T-cell responses and cytokine production in allergy patients, we have developed an in vitro system by using highly purified T cells as responder cells and monocyte-derived DC (MDC) as the APC. MDC provide a convenient and potent APC source for T-cell response studies.

Endocytosis by antigen presenting cells: dendritic cells are as endocytically active as other antigen presenting cells.

PMID: 1355605 | Indexed in PubMed

Although dendritic cells are the most potent of all antigen presenting cells, they have paradoxically been regarded as having only a minimal capacity for endocytosis, which is a crucial step in antigen processing prior to presentation. Previous studies of dendritic cells, which are only available in small numbers, have been restricted to measurement of long-term endocytosis and so have stressed lysosomal accumulation. Measurement of traffic through late endosomes, which are closely related to the organelle in which antigen processing occurs, has, to date, required large numbers of cells and therefore has not been possible for dendritic cells. To resolve the paradox for dendritic cells, we have developed a flow cytometric assay of fluid-phase endocytosis that assesses late endosomal traffic by kinetic analysis of exocytosis in small numbers of cells. Using this assay, we show that fluid-phase endocytosis--in particular, traffic through late endosomes--is as active in dendritic cells as in other antigen presenting cells.

Bone marrow-derived dendritic cells serve as potent adjuvants for peptide-based antitumor vaccines.

PMID: 9090785 | Indexed in PubMed

Dendritic cells (DCs) are considered the most effective antigen-presenting cells (APCs) for primary immune responses. Since presentation of antigens to the immune system by appropriate professional APCs is critical to elicit a strong immune reaction and DCs seem to be quantitatively and functionally defective in the tumor host, DCs hold great promise to improve cancer vaccines. Even though they are found in lymphoid organs, skin and mucosa, the difficulty of generating large numbers of DCs has been a major limitation for their use in vaccine studies. A simple method for obtaining DCs from mouse bone marrow cells cultured in the presence of GM-CSF + interleukin 4 is now available. In four different tumor models, mice injected with DCs grown in GM-CSF plus interleukin 4 and prepulsed with a cytotoxic T

lymphocyte-recognized tumor peptide epitope developed a specific cytotoxic T lymphocyte response and were protected against a subsequent tumor challenge with tumor cells expressing the relevant tumor antigen. Moreover, treatment of day 5-14 tumors with peptide-pulsed DCs resulted in sustained tumor regression in five different tumor models. These results suggest that presentation of tumor antigens to the immune system by professional APCs is a promising method to circumvent tumor-mediated immunosuppression and is the basis for ongoing clinical trials of cancer immunotherapy with tumor peptide-pulsed DCs.

Newly Found Rat CD103⁺ Dendritic Cells Represent a Highly Immunogenic Subpopulation of Type-2 Conventional Dendritic Cells, Corresponding to Known Dendritic Cell Subsets in Mice and Humans.

PMID: 39754477 | Indexed in PubMed

Dendritic cells (DCs), the primary antigen-presenting cells, have traditionally been identified by CD103 molecules in rats, whereas mouse and human DCs are identified by CD11c molecules. However, this history does not preclude the existence of CD103⁺ DCs in rats. To explore this possibility, we examined MHCII⁺ cells in rat spleen and thymus, identifying a novel population of CD103⁺-MHCII⁺-CD45R⁺-CD172a⁺ cells. These cells are negative for CD103 and B cell marker CD45R, but positive for the type-2 conventional DC (cDC2) marker CD172a. Transcriptomic analyses revealed that they represent a subpopulation of cDC2. Additionally, gene set enrichment analysis predicted enhanced immunogenic activities for this novel population compared to known rat cDC2s. Mixed leukocyte reaction assays confirmed that the rat CD103⁺ cDC2s induce T cell proliferation more effectively than other DC subsets, suggesting enhanced immunogenic potential. In reaggregated thymic organ culture assays, both the rat CD103⁺ and CD103⁺ cDC2 subsets suppressed the total number of generated thymocytes and skewed the differentiation toward CD8 single-positive cells. Comparisons with previously published single-cell RNA-sequencing datasets showed that the rat CD103⁺ cDC2 subset shares markers and GO terms of known mouse and human cDC2 subpopulations: cDC2a, cDC2b, inf-cDC2, and moDC. In contrast, the classic rat CD103⁺ cDC2 subset expresses only cDC2a markers. These findings provide new insights into DC subpopulations, particularly in species other than mice and humans, where much remains to be uncovered.

Current knowledge about interactions between avian dendritic cells and poultry pathogens.

PMID: 31830703 | Indexed in PubMed

In poultry production conditions today, birds are surrounded by viral, bacterial, and parasitic agents. DCs are the main antigen-presenting cells located in tissues of the body, and their role involves recognizing antigen structures, engulfing and processing them, and subsequently presenting antigen peptides on their surface by major histocompatibility complex, where T cells and B cells are stimulated and can begin appropriate cellular and antibody immune response. This unique function indicates that these cells can be used in producing vaccines, but first it is necessary to culture DCs *in vitro* to identify the principles of their interactions with pathogens. The following review summarizes our current knowledge about avian dendritic cells and their interactions with pathogens. It provides a basis for future studies of these unique

cells and their use in vaccine development.

Dendritic cells: tools and targets for antitumor vaccination.

PMID: 16221071 | Indexed in PubMed

Dendritic cells are the most potent antigen-presenting cells of the immune system and represent a promising tool in therapeutic vaccination against cancer. Immunotherapy applying ex vivo-generated and tumor antigen-loaded dendritic cells has been successfully introduced in clinical vaccination protocols and has proven to be feasible and effective in some patients. A better understanding of how dendritic cells succeed to induce and modulate immunity is necessary to optimally exploit dendritic cells in anticancer vaccines. The authors will review novel insights in antigen loading, activation and migration of dendritic cells and their impact on the application of ex vivo-generated dendritic cell vaccines. In addition, novel means to exploit dendritic cells in cancer vaccines by loading and activation of dendritic cells directly in situ and possible obstacles that should be overcome to induce long-lasting immunity in therapeutic settings will be discussed.

Characteristics of antigen-independent and antigen-dependent interaction of dendritic cells with CD4+ T cells.

PMID: 7545116 | Indexed in PubMed

Dendritic cells (DC) are the main antigen-presenting cells for the initiation of primary T cell-mediated immune responses. In the first stage of activation, T cells bind to DC in an antigen-independent manner. We studied the adhesion characteristics of human CD4+ T cells to DC generated from CD34+ hematopoietic progenitors following 12 to 13 days of culture in the presence of granulocyte/macrophage colony-stimulating factor and tumor necrosis factor- α . A majority of these cells had the morphology, phenotype and functions of DC. CD4+ T/DC adhesion was measured by means of fluorescence microscopy and flow cytometry. Four independent receptor/ligand pathways, LFA-1/ICAM, ICAM/LFA-1, CD2/LFA-3 and CD28/CD80, were involved in the transient adhesion of DC to CD4+ T cells in antigen-independent and specific alloantigen-dependent situations, as shown by blocking experiments using monoclonal antibodies. The antibodies also blocked a primary mixed lymphocyte reaction (MLR) in which DC were used as stimulatory cells. Adhesion of alloreactive CD4+ T cells to antigen-presenting DC was stronger than that of resting CD4+ T cells, while peak adhesion occurred after 5 and 20 min, respectively. The LFA-1 ligands involved in adhesion of resting CD4 T cells to DC and alloreactive CD4+ T cells to specific DC differed in part, since ICAM-3 on resting T cells and ICAM-1 on alloreactive T lymphocytes preferentially bound LFA-1. Studies of interactions between DC and phorbol ester-activated T cells expressing the CD40 ligand revealed a fifth independent adhesion pathway, CD40/CD40 ligand. CD4-mediated regulation of CD4+ T/DC adhesion was suggested by the observation that preincubation of CD4+ T cells and DC individually with anti-CD4 antibodies inhibited adhesion. In addition, antibodies specific for HLA class II molecules inhibited adhesion when used to pretreat DC but not alloactivated CD4+ T cells.

Mycobacterium tuberculosis GroEL2 Modulates Dendritic Cell Responses.

PMID: 29133346 | Indexed in PubMed

Mycobacterium tuberculosis successfully subverts the host immune response to promote disease progression. In addition to its known intracellular niche in macrophages, *M. tuberculosis* interferes with the functions of dendritic cells (DCs), which are the primary antigen-presenting cells of the immune system. We previously showed that *M. tuberculosis* dampens proinflammatory responses and impairs DC functions through the cell envelope-associated serine protease Hip1. Here we present data showing that *M. tuberculosis* GroEL2, a substrate of Hip1, modulates DC functions. The full-length GroEL2 protein elicited robust proinflammatory responses from DCs and promoted DC maturation and antigen presentation to T cells. In contrast, the cleaved form of GroEL2, which predominates in *M. tuberculosis*, was poorly immunostimulatory and was unable to promote DC maturation and antigen presentation. Moreover, DCs exposed to full-length, but not cleaved, GroEL2 induced strong antigen-specific gamma interferon (IFN- γ), interleukin-2 (IL-2), and IL-17A cytokine responses from CD4⁺ T cells. Moreover, the expression of cleaved GroEL2 in the *hip1* mutant restored the robust T cell responses to wild-type levels, suggesting that proteolytic cleavage of GroEL2 allows *M. tuberculosis* to prevent optimal DC-T cell cross talk during *M. tuberculosis* infection.
